Intelligent Traffic Management Using Fuzzy Logic and Machine Learning

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Final Report

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ABSTRACT

Urban traffic congestion has become a major issue in modern cities due to rapid urbanization, increasing population, and the continuous growth of vehicle ownership. Traditional traffic management systems, which rely on fixed-time signal controls and manual supervision, are often unable to respond effectively to changing traffic conditions. This study proposes an Intelligent Traffic Management System aimed at improving traffic flow efficiency, reducing congestion, and enhancing road safety through real-time monitoring and adaptive control techniques. The system integrates data from multiple sources such as IoT-based sensors, traffic cameras, and GPS-enabled devices to continuously monitor vehicle density, movement patterns, and intersection congestion levels. This real-time data is processed using machine learning algorithms to predict traffic conditions and dynamically adjust traffic signal timings based on current demand. By adapting to live traffic situations, the system helps optimize road usage and minimize unnecessary delays at intersections. In addition, the proposed system includes emergency vehicle prioritization, ensuring that ambulances, fire trucks, and other emergency services receive immediate green signals to improve response times. A mobile application is also incorporated to provide drivers with real-time traffic updates and alternative route suggestions, helping them avoid congested areas. This leads to reduced travel time, fuel consumption, and environmental pollution. Prototype implementation demonstrates that the intelligent system significantly outperforms conventional fixed-timing traffic control methods in terms of efficiency and responsiveness. The results show improved traffic flow, reduced congestion, and better overall commuter experience. Furthermore, the system is scalable and can be integrated into existing urban infrastructure, making it suitable for future smart city developments. This research emphasizes the importance of combining IoT, artificial intelligence, and data analytics to develop efficient and adaptive traffic management solutions for modern urban environments.

Keywords: Intelligent Traffic Management, Smart City, IoT Sensors, Machine Learning, Traffic Congestion, Real-time Data Analytics, Adaptive Signal Control

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Contents

ABSTRACT ..... **iii**

ACKNOWLEDGEMENT ..... **iv**

LIST OF TABLES ..... **vi**

LIST OF FIGURES ..... **vi**

LIST OF ABBREVIATIONS ..... **vii**

1. INTRODUCTION **1**

- Parking Violation Detection & Management 2
- Traffic Violation Detection & LiveSafe Score 3
- Fuzzy Logic Adaptive Traffic Signal Controller 3
- Accident Risk Prediction and Integrated Dashboard 3

1.1 Background and Literature Survey **5**

1.2 Research Gap **7**

1.3 Research Problem **14**

1.4 Research Objectives **17**

1.4.1 Main Objective 17

1.4.2 Sub Objective 17

2. METHODOLOGY **18**

2.1 Commercialization aspects of the product **23**

2.2 Testing & Implementation **24**

2.2.1 Implementation 24

2.2.2 Testing 25

2.2.3 Integration and System Testing 25

3. RESULTS & DISCUSSION **26**

3.1 Results **26**

3.1.1 IoT-Based Traffic Monitoring Results 26

3.1.2 Machine Learning Traffic Prediction Results 27

3.1.3 Adaptive Traffic Signal Control Results 27

3.1.4 Route Optimization System Results 27

3.1.5 Accident Risk Prediction System Results 28

3.2 Research Findings and Discussion 28

3.2.1 Effectiveness of IoT-Based Traffic Monitoring 28

3.2.2 Impact of Machine Learning in Traffic Prediction 28

3.2.4 Contribution of Route Optimization System 29

3.2.5 Effectiveness of Accident Risk Prediction System 29

3.2.6 Overall System Integration Analysis 30

3.3 Summary of Each Student's contribution 30

4. CONCLUSION 35

5. REFERENCES 37

LIST OF TABLES

Table 1.2.2: Comparison of Research Gaps Across Existing Systems **13**

LIST OF FIGURES

Figure 2.1 Overall Component System Diagram **19**

Figure 2.2 YOLOv8n Object Detection Process **20**

Figure 2.3 Parking Violation Detection Flow **21**

Figure 2.4 Traffic Violation Detection and Safety Score Workflow **21**

Figure 2.5 Adaptive Traffic Signal Control Workflow **22**

Figure 2.6 Accident Risk Prediction Workflow **22**

LIST OF ABBREVIATIONS

Abbreviation	Description
ML	Machine Learning
FL	Fuzzy Logic
RFID	Radio-Frequency Identification
AI	Artificial Intelligence
ITMS	Intelligent Traffic Management System.
IR	Infrared
EVP	Emergency Vehicle Prioritization
MCU	Microcontroller Unit (ESP32)
IOT	Internet of Things
YOLO	You Only Look Once
DE	Differential Evolution
MQTT	Message Queuing Telemetry Transport
AWS	Amazon Web Services
V2I	Vehicle-to-Infrastructure
OBU	Onboard Unit
LED	Light-Emitting Diode
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
OCR	Optical Character Recognition
RU	Roadside Unit
GPS	Global Positioning System
CCTV	Closed-Circuit Television

1. INTRODUCTION

Rapid urbanization and the continuous growth in vehicle ownership have significantly intensified traffic-related challenges in Sri Lanka. Major urban areas experience severe congestion, illegal parking, frequent traffic violations, and an increasing number of road accidents. These issues not only disrupt daily transportation but also negatively impact economic productivity, environmental sustainability, and public safety. Efficient traffic management has therefore become a critical requirement for ensuring smooth mobility and minimizing delays in modern urban environments.

Traditional traffic management systems in Sri Lanka primarily rely on fixed-time traffic signals and manual traffic monitoring. While these approaches were effective in earlier, less complex traffic conditions, they are no longer sufficient to handle the dynamic and unpredictable nature of modern traffic flows. Fixed-time signal systems operate based on predefined schedules without considering real-time traffic density, resulting in inefficient utilization of road infrastructure. Similarly, manual monitoring and enforcement are labor-intensive and often delayed, reducing their effectiveness in preventing violations and congestion. Studies have shown that traditional traffic signal systems lack adaptability and fail to respond efficiently to real-time traffic variations [1]. Consequently, traffic congestion and unsafe driving behaviors frequently remain unaddressed until accidents or serious violations occur.

In recent years, advancements in intelligent systems, computer vision, and data analytics have enabled the development of smart traffic management solutions. Deep learning-based models such as YOLOv8 have significantly improved the accuracy and efficiency of real-time vehicle detection and tracking. Additionally, the integration of sensors and Internet of Things (IoT) devices allows continuous data collection from traffic environments, enabling real-time monitoring and analysis. Intelligent Traffic Management Systems (ITMS) leverage these technologies to optimize traffic flow, enhance road safety, and support informed decision-making through continuous data processing [2].

Despite the availability of traffic cameras and basic monitoring infrastructure, current traffic management practices in Sri Lanka face several limitations. Illegal roadside parking is often identified only after congestion has already occurred, leading to delayed enforcement actions. Traffic violations, including speeding, red-light violations, and improper lane usage, are largely

detected through manual review processes, which are time-consuming and inefficient. Furthermore, fixed-time traffic signal systems are incapable of adapting to fluctuating traffic volumes and do not provide effective prioritization for emergency vehicles. Accident prevention mechanisms are also predominantly reactive, with limited capabilities for early risk detection. Moreover, the lack of integration among different traffic management components results in poor situation awareness and ineffective decision-making. These challenges highlight the need for advanced, integrated solutions that combine real-time monitoring, predictive analytics, and adaptive control mechanisms [3].

To address these challenges, this research proposes an **Intelligent Traffic Management System (ITMS)** that integrates fuzzy logic and machine learning techniques into a unified framework tailored for Sri Lankan traffic conditions. The proposed system is designed to provide real-time monitoring, predictive enforcement, adaptive signal control, and comprehensive risk assessment. By combining multiple intelligent components, the system aims to overcome the limitations of existing approaches and deliver a more efficient and proactive traffic management solution.

The proposed ITMS consists of four key components, each targeting a critical aspect of traffic management,

- **Parking Violation Detection & Management**

Illegal parking is a major contributor to traffic congestion in urban areas. Vehicles parked in restricted zones reduce road capacity, obstruct traffic flow, and increase accident risks. The proposed system introduces a Parking Behavior Analysis module that utilizes computer vision techniques to detect vehicles, monitor their movement, and determine whether they are stationary within restricted zones.

The system tracks parking duration and differentiates between temporary stops and actual violations. It also incorporates predictive warning mechanisms that notify drivers before a violation is confirmed. A dynamic fine calculation model is implemented to ensure fairness by considering factors such as violation duration and traffic impact. This approach enhances enforcement efficiency while encouraging responsible parking behavior.

- ## Traffic Violation Detection & LiveSafe Score

Traffic violations play a significant role in road accidents and unsafe driving conditions. The proposed system includes a Smart Traffic Violation Detection module capable of identifying violations such as speeding, red-light running, lane weaving, and wrong-way driving using real-time video analysis.

In addition to detecting violations, the system introduces a **LiveSafe Score**, a safety metric that evaluates the overall safety level of a traffic junction. This score is dynamically updated based on detected violations and contextual factors such as traffic density. By providing real-time safety information, the system promotes community awareness and encourages safer driving behavior rather than focusing solely on punitive measures.

- ## Fuzzy Logic Adaptive Traffic Signal Controller

Conventional traffic signal systems are unable to respond effectively to dynamic traffic conditions. To address this limitation, the proposed system incorporates a **Fuzzy Logic–based Adaptive Traffic Signal Controller**. Fuzzy logic is well-suited for traffic management as it can handle uncertainty and imprecise data using linguistic variables such as "low," "medium," and "high" traffic density.

The controller dynamically adjusts signal timings based on real-time vehicle counts, optimizing traffic flow at intersections. It also supports emergency vehicle prioritization by granting immediate right-of-way when necessary. This adaptive mechanism significantly reduces congestion and improves intersection efficiency compared to fixed-time systems.

- ## Accident Risk Prediction and Integrated Dashboard

Road accidents are often influenced by multiple factors, including driving behavior, traffic violations, and speed patterns. The proposed system includes an **Accident Risk Prediction module** that analyzes these factors to estimate the likelihood of potential accidents in real-time.

The system calculates a risk score for each vehicle based on its behavior and historical data, enabling early identification of high-risk situations. This predictive capability allows for proactive interventions, reducing the likelihood of accidents. Additionally, an integrated dashboard provides

real-time visualization of traffic conditions, violations, risk levels, and system analytics. This dashboard serves as a decision-support tool for traffic authorities, enhancing situational awareness and enabling data-driven decision-making.

The primary contribution of this research lies in the integration of multiple traffic management functionalities into a single unified system. Unlike existing solutions that address individual problems in isolation, the proposed ITMS combines detection, control, and prediction mechanisms within a real-time framework. This integrated approach enables:

- Real-time traffic monitoring and automated violation detection
- Adaptive traffic signal control using fuzzy logic
- Predictive accident risk assessment using machine learning
- Context-aware safety evaluation through the LiveSafe Score
- Improved enforcement efficiency and driver behavior awareness

By addressing both operational efficiency and road safety, the system provides a comprehensive solution for modern traffic management challenges.

The remainder of this thesis is structured as follows. Section II reviews the relevant literature related to intelligent traffic management, computer vision, and IoT-based solutions. Section III describes the proposed system architecture and methodology. Section IV presents the implementation details and experimental results. Finally, Section V concludes the study and outlines potential future improvements.

In conclusion, the increasing complexity of urban traffic systems necessitates intelligent, adaptive, and integrated solutions. The proposed Intelligent Traffic Management System represents a significant advancement by combining fuzzy logic and machine learning within a unified framework tailored to Sri Lankan conditions. Through its four key components—parking management, violation detection, adaptive signal control, and accident risk prediction—the system aims to enhance traffic efficiency, improve road safety, and support proactive traffic management strategies.

1.1 Background and Literature Survey

The evolution of urban mobility management has moved from static signaling to complex computational frameworks. Early attempts to alleviate traffic congestion frequently depended on Multi-Agent systems, which used localized "Controller Agents" to regulate automobile and pedestrian traffic. However, as Wakkumbura et al. (2021) [4] point out, these systems are usually hampered by rigid, rule-based logic that fails to adapt to the fluid nature of real-world traffic. While agent-centric models lay the groundwork for decentralized control, they rely significantly on exact density data, which is frequently unavailable in chaotic urban contexts. In contrast, the Intelligent Traffic Management System (ITMS) incorporates fuzzy logic to provide a context-aware solution. This enables the system to handle "uncertain and poor" real-time data by transforming sensor inputs into dynamic signal alterations that exceed classic request-based models.

The basic principles constraint of traditional traffic modeling is its reactive nature and reliance on past data. Thathsarani and Lanel [5] used graph theory and simulation models to discover Colombo's critically crowded highways and parking zones. While such models are useful for urban planning and identifying historical bottlenecks, they do not provide the active, AI-powered detection required for real-time intervention. The ITMS bridges this technology gap by combining a proprietary YOLOv8 model with DeepSORT tracking. Unlike static simulation-driven models, this approach allows for real-time classification of roadside activities, including a 15-minute grace period and mobile driver alerts. The ITMS ensures a smoother urban flow than previous modeling approaches by identifying between legal and illegal stops using Roadside Parking Behavior Analysis.

Historically, traffic enforcement techniques used Optical Character Recognition (OCR) to automate the recognition of license plates. The system created by Conara et al. [6] demonstrates the efficacy of database-driven enforcement for registration and infraction tracking. However, such systems use binary logic either a violation occurred or it did not while neglecting the environmental context and the actual consequence of the transgression on road safety. The ITMS goes beyond this binary approach by establishing the LiveSafe Score System. This context-aware module assesses the "impact" of a violation depending on environmental factors such as traffic congestion and time of day.The system creates a real-time safety score using a

severity weighting algorithm, guaranteeing that enforcement is more than just penalties; it also has about known the risk level of specific behaviors in various settings.

Navigation-based safety interventions have also been investigated to prevent accidents by changing driver behavior. Jayamanna and Kalansooriya [7] developed intelligent navigation software that notifies drivers to accident hotspots using historical data. While this raises driver awareness of high-risk areas, they are fundamentally restricted by their dependence on past accidents. The ITMS marks a proactive shift in transportation safety. Rather of telling a driver where an accident happened, the system optimizes real-time traffic flow to avoid the conditions that cause accidents in the first place.Authorities may monitor and reduce risks before they become crashes by using the ITMS's interactive dashboard, which generates dynamic risk scores based on vehicle speed, inter vehicle distance, and current infractions.

Layered designs for accident detection have been developed via the integration of the Internet of Things (IoT), frequently using Roadside Units (RSUs) and clustering algorithms. A complex IoT-based adaptive traffic management system with a focus on infrastructure and backend governance was proposed by Lilhore et al. [8]. However, efficient management on the road is not always correlated with technical sophistication in the backend. The single, user-focused interface required for proactive decision-making is absent from many of these systems. By integrating complex logic such as fuzzy-logic risk assessments and adaptive signaling into a single platform for human-machine interaction, the ITMS enhances existing infrastructure-heavy models. This prioritizes accessibility and localized machine learning, guaranteeing that the system is a dependable, safety-focused solution tailored to modern urban mobility rather than simply a passive monitor.

One of the major advantages of the ITMS over existing multi-agent and OCR-based systems is its capacity to deal with "unruly" traffic patterns found in expanding urban areas. While international algorithms frequently fail to distinguish a variety of vehicle types, such as tuk-tuks or overcrowded buses, the ITMS's unique machine learning architecture is trained specifically for these local settings. Traditional multi-agent frameworks [4] frequently presume lane discipline and vehicle uniformity, which are not common in many high-density areas. By leveraging customized computer vision, the ITMS bridges the gap between high-level theory and the practical reality of uncertain road use, giving a more robust detection capability than

earlier research systems [4][6].This adaptability ensures that the system functions properly even when traffic behavior deviates from established rules.

Furthermore, the shift from passive automation to empowered human-machine interaction is a watershed moment in traffic management research. While systems like those described by Lilhore et al. [8] provide the necessary IoT foundation, they frequently fail to provide actionable insights to real-time authorities. The ITMS solves this by visualizing data via live video feeds and risk assessments, thereby transforming raw sensor data into strategic knowledge. This technique lightens the cognitive load on traffic officials, allowing for more exact adjudication and intervention. The integration of community comments and IoT notifications promotes a collaborative safety environment, which differs significantly from the solely punitive systems present in previous registration-based detection models [6].

Finally, the Intelligent Traffic Management System (ITMS) is a full creation of fuzzy logic, deep learning, and real time monitoring. Analyzing the constraints of multi-agent systems [4], historical simulations [5], binary enforcement models [6], reactive safety apps [7], and infrastructure-heavy IoT designs [8] demonstrates that a mixed, proactive strategy is required for modern urban contexts. The ITMS not only detects congestion, but also actively attempts to alleviate it using adaptive signal regulation and behavioral analysis. This system is the most modern, adaptable, and safety-oriented solution available today, providing a level of optimization and real-time decision-making power that much exceeds that of standard traffic management frameworks.

1.2 Research Gap

In South Asia, the shift from manual, traditional traffic monitoring to automated intelligent systems is still inadequate, especially when it comes to integrating adaptive infrastructure, real-time behavioral analysis, and complete driver accountability. Instead of addressing traffic management as a cohesive, data-driven ecosystem, current traffic management approaches in Sri Lanka, particularly in the Colombo urban area, usually address traffic management as a collection of isolated problems. Although a number of international studies have suggested intelligent traffic solutions, a critical review of the literature indicates that no single system has addressed the entire range of issues, such as proactive accident risk prediction, adaptive signal control, traffic violation monitoring with driver scoring, and illegal parking detection, within a

unified framework specific to developing country contexts. The limits of current research are methodically examined in the following sections, which also show how the suggested Intelligent Traffic Management System (ITMS) fills up these gaps.

1.2.1 Limitation of Existing Traffic Management Approaches

Using algorithmic evolution to optimize the Random Early Detection for Vehicles Dynamic (iREDVD) mechanism, Guillen-Perez and Cano [9] created an intelligent Internet of Things-based traffic management system. In comparison to conventional fixed-time systems, their work showed significant improvements in vehicle waiting time reduction of about 27% as well as savings in emissions and fuel usage. However, this system does not handle parking behavior analysis, traffic violation detection, or accident risk prediction instead, it only concentrates on adaptive signal control at isolated crossings. Although effective for signal improvement, the iREDVD mechanism lacks a driver accountability system and the ability to trace infractions. On the other hand, the suggested ITMS combines a thorough violation detection pipeline, a driver safety scoring system, and adaptive signal regulation utilizing fuzzy logic. Although Guillen-Perez and Cano [9] achieved significant gains in signal timing, their system is unable to detect illegal parking, identify red-light violations, or give drivers real-time risk assessments. This represents a significant gap that the proposed ITMS fills by combining several intelligent modules into a single architecture.

Amarnath and colleagues [10] introduced the Intelli Traffic Management System, which uses a Priority Round Robin Algorithm for adaptive signal control, LSTM neural networks for congestion prediction, and YOLO v11 for vehicle recognition. For traffic administrators, its web-based interface offers both manual override capabilities and real-time traffic visualization. Nevertheless, this approach has a number of significant drawbacks. Initially the process for detecting violations is characterized as basic detection only, with no systematic method for identifying and recording particular infractions like speeding, red-light violations, or lane-weaving. Furthermore, neither a longitudinal driver behavior tracking system nor a driver safety score exist. Next, their architecture makes no mention of parking infraction management. Finally, instead of using real-time risk score based on current vehicle speed, inter-vehicle distance, and violation history, accident risk prediction uses LSTM-based congestion forecasting. A comprehensive Traffic Violation Detection system with automated evidence

capture, a Driver Safety Scoreboard that maintains daily recalculated scores based on violation severity, an Accident Risk Prediction module that computes dynamic risk scores combining instantaneous speed factors with historical behavioral factors, and a dedicated Parking Violation Management module with a 30-second grace period and visual-audible warnings are all included in the proposed ITMS to address each of these gaps.

Baquero and colleagues performed a bibliometric investigation about the utilization of Artificial Intelligence, Big Data, and the Internet of Things in traffic management systems, focusing primarily on Latin America. Despite having multiple developed studies, their research showed that Ibero-America exhibits minimal importance inside the global field of study. Critical issues were noted by the study, such as inadequate systems, high implementation costs, reluctance to change, a lack of infrastructure for real-time data, and challenges in gathering user data. Most significantly, the study discovered that nations like Brazil, Mexico, and Peru have suggested cheap replacements that depend on smartphone-based IoT connectivity instead of extensive infrastructure improvements. Nevertheless, no detecting algorithm, predictive model, or technical solution is suggested by this bibliometric investigation. It doesn't fill the holes; it just finds them. By creating a system specifically for Colombo, Sri Lanka, using a localized YOLOv8 training dataset that contains region-specific vehicle types including buses and three-wheelers, the suggested ITMS directly solves the lack of region-specific implementation. The system is appropriate for poor nation contexts where budget restrictions are severe since it uses affordable ultrasonic and infrared sensors instead of costly infrastructure.

Islam [12] integrated results from 68 empirical research and government performance records from 2010 to 2024 in a meta-analysis of AI-enabled Adaptive Traffic Control Systems for urban mobility optimization. In comparison to pre-implementation baselines, the investigation showed that AI-driven solutions lower average vehicle delay by 24% to 36%, intersection queuing by 28%, and total trip time by up to 19%. Deep Reinforcement with Multiple Agents Under dynamic flow conditions, learning techniques demonstrated improved scalability. However, this meta-analysis does not address driver behavior change or traffic infraction detection; instead, it solely concentrates on traffic flow optimization. The systems examined by Islam [12] do not distinguish between safe and unsafe drivers; instead, all vehicles are treated as equal agents, and there is no way to reward compliance or penalize risky behavior.

The proposed ITMS directly addresses this gap with the Driver Safety Scoreboard, which gives each driver a starting score of 100 and subtract points based on the severity and frequency of identified violations. This technique generates a longitudinal record of driver behavior, allowing for tailored interventions for high-risk drivers. In contrast to Islam's [12] analysis of historical accident data, the proposed ITMS uses Equation (2) to predict real-time accident risk: $S_{risk} = (F_{speed} \times 0.6) + (F_{history} \times 0.4)$, where F_{speed} represents instantaneous speed-related risk and $F_{history}$ represents behavioral history from violation database. When risk ratings approach CRITICAL levels, automatic alarms appear on the dashboard, allowing for proactive intervention before an accident occurs.

None of the reviewed systems offer automated parking violation management, including duration tracking and predictive deterrence. The existing literature regards parking as a separate issue from traffic flow management. Guillen-Perez and Cano [9] focus solely on signal time optimization for crossings with no parking monitoring capability. Amarnath and colleagues [10] did not mention parking infraction detection in their Intelli Traffic Management System. Baquero and colleagues [11] acknowledge infrastructure issues but offer no parking solutions. Islam [12] examines traffic flow metrics without considering parking behavior. This is a serious gap since unlawful roadside parking causes significant congestion on narrow Sri Lankan roadways like Galle Road and Duplication Road in Colombo. The proposed ITMS covers this gap with a specific parking violation. The proposed ITMS addresses this need by including a specialized Parking Violation Management module that monitors vehicles in restricted zones, tracks stop length, and records supporting data for vehicles that remain parked. To calculate dynamic fines, use the formula $F_{total} = [F_{base} + (T_{duration} \times R_{time})] \times I_{impact}$. Penalties increase proportionally with longer parking periods and higher traffic impact. This proactive barrier method changes the emphasis from punitive enforcement to congestion prevention.

All examined systems lack a complete dual-interface environment that caters to both drivers and traffic authority. Guillen-Perez and Cano [9] offer no driver-facing applications. Amarnath and colleagues [10] provide a web-based dashboard for administrators, but no mobile app for drivers to monitor violations or pay fines. Baquero and colleagues [11] examine existing systems but offer no connection. Islam [12] examines urban mobility indicators without considering user interfaces. This divide is crucial since traffic management necessitates contact

with both law enforcement and individual drivers. The proposed ITMS addresses this by utilizing a dual-interface environment. A Driver Mobile Application that enables drivers to access their Safety Score, examine recorded violations, and pay penalties, as well as a Command Center Dashboard for authorities to monitor live video feeds, violation notifications, responsive signal control, and risk heatmaps. This connection translates complex backend logic into understandable visualizations, allowing for speedier decision-making and more transparent communication between citizens and law enforcement organizations.

1.2.2 Comparison of Research Gaps Across Existing Systems

The table below presents a comprehensive comparison of features and research gaps in the examined literature with the proposed ITMS.

Feature / Domain	**Guillen-Perez and Cano [9]**	**Amarnath et al. [10]**	**Baquero et al. [11]**	**Islam [12]**	**Proposed ITMS**
Primary Technology	IoT and Evolutionary Algorithms.	YOLO v11 and LSTM	Bibliometric Review (AI/IoT)	AI-Adaptive Control (Meta-analysis)	Fuzzy Logic + YOLOv8 + IoT
Domain Focus	Optimization of Signals	Detection & Control	Regional Analysis	Urban Mobility	Management of Unified Urban Roads
Parking Violation Detection	Not present	Not present	Not present	Not present	Comprehensive parking management application.
Red-Light Violation Detection	Not present	Basic detection	Not present	Not present	Automated detection of evidence

Over-Speeding Detection	Not present	Not specified	Not present	Not present	Real-time speed measuring
Lane-Weaving Detection	Not present	Not specified	Not present	Not present	Detected using severity weighting
Driver Safety Score	Not present	Not present	Not present	Not present	Daily revised score (0-100)
Driver Mobile Application	Not present	Not present	Not present	Not present	Viewing the violation and paying the fine.
Accident Risk Prediction	Not present	LSTM congestion only	Not present	Historical meta-data only	Real-time Srisk = (Fspeed × 0.6) + (Fhistory × 0.4)
Risk Heatmap Visualization	Not present	Basic dashboard	Not present	Not present	Heatmaps are interactive and real-time.
Adaptive Signal Control	iREDVD mechanism	Priority Round Robin	Not present	RL-based (analyzed)	RFID preemption in fuzzy logic
Emergency Vehicle Prioritization	Not present	Not specified	Not present	Not specified	RFID-based override
IoT Sensor Integration	Ultrasonic, IR	IoT cameras	Theoretically discussed	General sensors	IR + RFID + CCTV

Real-Time Violation Alerts	Not present	Not specified	Not present	Not present	Mobile notifications
Command Center Dashboard	Basic visualization	Web-based	Not present	Performance metrics	Integrated dashboard featuring risks, signals, infractions, and live feeds
Dynamic Fine Calculation	Not present	Not specified	Not present	Not applicable	Ftotal = [Fbase + (Tduration × Rtime)] × Iimpact
Multi-Object Tracking	Not present	Not specified	Not present	Not specified	BoT-SORT for effective vehicle identification.

Table 1.2.2: Comparison of Research Gaps Across Existing Systems

According to the comparative analysis, the suggested ITMS fills in a number of important research gaps left by current systems. Although Guillen-Perez and Cano [9] were able to optimize the signal, their system does not have a means for detecting infractions or holding drivers accountable. Despite implementing vehicle detection, Amarnath and colleagues' system does not include driver safety scoring or parking violation management [10]. While identifying regional issues, Baquero and colleagues [11] did not offer any technical solutions. Islam [12] examined the advantages of adaptive control but did not address real-time danger prediction or driver behavior adjustment. A dual interface environment that serves both drivers and authorities is not offered by any of the current systems.

All of the evaluated research lacks parking behavior analysis with duration tracking and predictive deterrence. No system now in use uses real-time accident risk rating that combines previous behavioral characteristics with instantaneous speed factors. With localized YOLOv8

training on three-wheelers and buses, fuzzy logic adaptive control with RFID emergency preemption, driver safety scoring, a mobile application for fine payment, and real-time risk prediction producing interactive heatmap visualizations, the proposed ITMS uniquely integrates all these missing components within a unified, modular architecture specifically designed for Sri Lankan traffic conditions.

1.3 Research Problem

The fast urbanization and increasing vehicle density in Colombo, Sri Lanka, have generated an urgent need for efficient traffic control systems. Traditional traffic management systems in the country continue to rely on fixed-time signals and manual monitoring, which are ineffective in responding to real-time traffic changes. While various foreign studies have presented advanced traffic management frameworks, a rigorous assessment demonstrates that existing methods have fundamental shortcomings when used in poor countries such as Sri Lanka. The research problem addressed in this study is the lack of an integrated, context-aware ITMS that can handle parking behavior analysis, traffic violation detection with driver accountability, adaptive signal control using FL, and proactive accident risk prediction all within a unified, lightweight architecture suitable for resource constrained urban settings.

Milanes and colleagues [13] created an innovative Vehicle-to-Infrastructure (V2I) traffic control system that operates by direct connection between vehicles and roadside infrastructure. Their approach requires that each vehicle be equipped with an Onboard Unit (OBU) capable of reporting speed, position, and destination data to a central control station, which then computes optimal speed recommendations and provides warning signals to drivers. The technology improved traffic flow and prevention by using a fuzzy-based control algorithm to calculate reference speeds and safe spacing between vehicles. However, this V2I-based technique has some significant shortcomings that render it inappropriate for Sri Lankan settings. The demand that every vehicle have OBU hardware and active communication capabilities is impossible to achieve in Colombo, since the vast majority of vehicles, including three-wheelers, older model cars, and motorcycles, lack any sort of digital communication technology. Milanes and colleagues [13] presented a system that assumes all drivers will obey the recommended speed and distance guidelines, with no method to identify or penalize drivers who violate traffic rules by exceeding the speed limit, running red lights, or illegally parking. The control station in their system can only analyze information from communicating vehicles

and cannot detect non-communicating vehicles, pedestrians, or stationary barriers such as illegally parked cars.

Furthermore, there is no driver accountability mechanism, no safety score tracking, and no mobile app that allows drivers to view their driving behavior or get violation notices. The proposed ITMS directly addresses all of these gaps by using passive computer vision with YOLOv8, which requires no vehicle modifications, detects all vehicles regardless of communication capability, automatically identifies over speeding, red-light violations, and lane-weaving through video analysis, includes a dedicated Parking Violation Management module with 30-second grace period warnings, and provides a Driver Safety Scoreboard with a mobile application for fine payment.

Alam [14] did a thorough literature study on road traffic congestion prediction using AI and ML with spatiotemporal data. The review looked at several predictive modeling techniques, including supervised learning, deep learning with Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), reinforcement learning, anomaly detection, and short- and long-term forecasting models. The study found that deep learning models, particularly Long Short-Term Memory (LSTM) networks, outperformed other methods at capturing temporal dependencies and geographical correlations in traffic data. However, Alam's assessment [14] reveals some chronic issues that remain unaddressed. Data quality and availability challenges continue to impede model efficacy, with inconsistent, incomplete, or noisy data considerably lowering forecast accuracy.

Model understanding and transparency remain major obstacles, as deep learning models function as "black boxes," providing little understanding into how forecasts are created. The scalability of AI models to large-scale urban networks remains a challenge, with most studies focused on individual corridors rather than full citywide systems. Privacy concerns about collecting and processing spatiotemporal data from individual automobiles and personal devices have not been sufficiently addressed. Most importantly, Alam's evaluation [14] does not address traffic violation detection, driver behavior change, parking management, or emergency vehicle prioritizing, as the emphasis focuses only on congestion prediction rather than general traffic management. Using inexpensive IR sensors and passive Radio-Frequency Identification (RFID) in place of costly GPS-based tracking that raises privacy concerns,

concentrating on the Colombo urban area with localized YOLOv8 training data that includes three-wheelers, Ambulances, buses and combining congestion prediction with violation detection, parking monitoring, and emergency vehicle preemption within a single unified architecture, the proposed ITMS fills each of these gaps.

The total lack of driver accountability measures in current intelligent traffic systems is a major research issue that is clear from both Milanes and colleagues [13] and Alam [14]. Drivers who choose to disregard speed suggestions or break traffic laws face no repercussions under the V2I system put forth by Milanes and colleagues [13]. All cars are treated as equal agents that willingly send data and obey commands. The control station is unable to discern between a risky motorist who routinely exceeds speed limits or runs red lights and a safe driver who complies with all regulations.

Congestion prediction models that concentrate on traffic flow patterns without taking into account individual driver behavior, infraction history, or safety scoring are examined in Alam's literature review [14]. Because traffic infractions are signs of long-term dangerous conduct patterns rather than isolated instances, this gap is crucial. Even while current systems treat all drivers equally, a driver who regularly violates Galle Road's lane-weaving laws during high-density rush hours poses a much higher danger of an accident than a driver who commits a single minor infraction during low-traffic conditions. In order to solve this, a motorist Safety Scoreboard that gives each registered motorist a starting score of 100 has been included in the proposed ITMS. Points are subtracted in accordance with a severity-weighting method that takes into account contextual aspects like traffic density and time of day when a violation exists using YOLOv8-based video analysis. For insurance purposes and focused enforcement actions, the daily recalculated score guarantees stability over time and offers a trustworthy signal of driver risk levels.

Parking violation management and emergency vehicle prioritizing are two more important research issues that are not included in current studies. Critical delays result from Milanes and colleagues' [13] system's whole attention on moving vehicles, which ignores parked cars and roadside stopping behavior. Additionally, emergency vehicles, such ambulances, are not given special treatment at intersections. Alam [14] examines algorithms for predicting traffic but ignores parking habits and emergency preemption. The proposed ITMS closes these gaps with

a special Parking Violation Management module that measures parking duration, employs YOLOv8-based computer vision to identify stopped vehicles in restricted zones, turns on warning LEDs following a 30-second grace period, and automatically produces dynamic fines. Furthermore, when an ambulance or fire truck approaches, the fuzzy logic controller may safely stop cross-traffic until the emergency vehicle passes thanks to an Emergency Vehicle Prioritization module that uses passive RFID technology.

In terms of sustainable accessibility and local fit, current systems show major research gaps. Alam [14] ignores user-centric design for undeveloped countries, while Milanes et al. [13] rely on specialized in-car technology without driver interfaces. To solve this, the suggested ITMS offers a dual-interface environment that consists of a Command Center Dashboard for real time authority monitoring and a Driver Mobile Application for Safety Score tracking and fine payments. Importantly, ITMS is tailored for Sri Lankan infrastructure, whereas current frameworks presume flawless compliance. To ensure accuracy in Colombo's particular environment, it makes use of inexpensive infrared sensors, passive RFID, and a YOLOv8 model that was trained on local Ambulances, three-wheelers and buses.

1.4 Research Objectives

1.4.1 Main Objective

The main objective of this research is to design, develop, and evaluate an **Intelligent Traffic Management System (ITMS)** that integrates fuzzy logic and machine learning techniques to enable real-time traffic monitoring, adaptive traffic control, automated violation detection, and accident risk prediction, with the aim of improving traffic flow efficiency and enhancing road safety in Sri Lankan urban environments.

1.4.2 Sub Objective

To achieve the main objective, the following specific objectives are defined:

1. **To develop an automated parking violation detection and management system** that accurately identifies illegally parked vehicles in restricted zones using computer vision techniques, monitors parking duration, and applies a dynamic fine calculation model based on violation duration and traffic impact.
2. **To design and implement a real-time traffic violation detection system** capable of detecting unsafe driving behaviors such as speeding, red-light violations, lane weaving, and wrong-way driving with high accuracy and minimal processing delay.

3. **To introduce a LiveSafe Score mechanism for junction-level safety evaluation** that dynamically assesses and represents the safety condition of traffic intersections based on detected violations and contextual traffic density.
4. **To implement a fuzzy logic–based adaptive traffic signal controller** that dynamically adjusts signal timings according to real-time traffic conditions and provides effective prioritization for emergency vehicles.
5. **To develop an accident risk prediction model** that evaluates the likelihood of potential accidents by analyzing vehicle speed patterns and historical violation data and categorizes vehicles into different risk levels.
6. **To design and implement an integrated real-time dashboard system** that visualizes traffic conditions, detects violations, safety scores, and risk levels to support informed decision-making by traffic authorities.
7. **To integrate all system components into a unified architecture** that ensures seamless data flow and real-time communication between detection, analysis, control, and visualization modules.
8. **To evaluate the performance and effectiveness of the proposed system** by measuring detection accuracy, system response time, reliability, and its impact on traffic flow and road safety under simulated or controlled conditions.

2. METHODOLOGY

A. System Overview and Methodology

This study proposes an integrated Intelligent Traffic Management System (ITMS) designed to solve major urban traffic problems such as congestion, unsafe driving behavior, traffic violations, and inefficient signal timing. The system combines several intelligent modules into one unified platform to support real-time traffic monitoring, improved traffic flow, and enhanced road safety.

At the center of the system is a shared data acquisition layer, which continuously collects real-time data from cameras, sensors, and IoT devices deployed in the traffic environment. This layer acts as the main communication backbone, ensuring that all system components receive synchronized and accurate data.

The collected data is distributed into four main processing modules: (1) Parking Violation Management, which detects illegally parked vehicles and monitors parking behavior; (2) Traffic Violation Detection, which identifies unsafe driving actions such as speeding and red-light

violations; (3) Adaptive Signal Control, which dynamically adjusts traffic signal timings based on traffic density; and (4) Accident Risk Prediction, which analyzes traffic patterns and driver behavior to predict potential accidents and generate early warnings.

By using this shared data layer, all modules operate efficiently while maintaining independence. This modular structure improves system scalability, reliability, and ease of maintenance, and allows future enhancements without modifying the entire system. The overall system architecture and interaction between these modules are illustrated in Fig. 1.

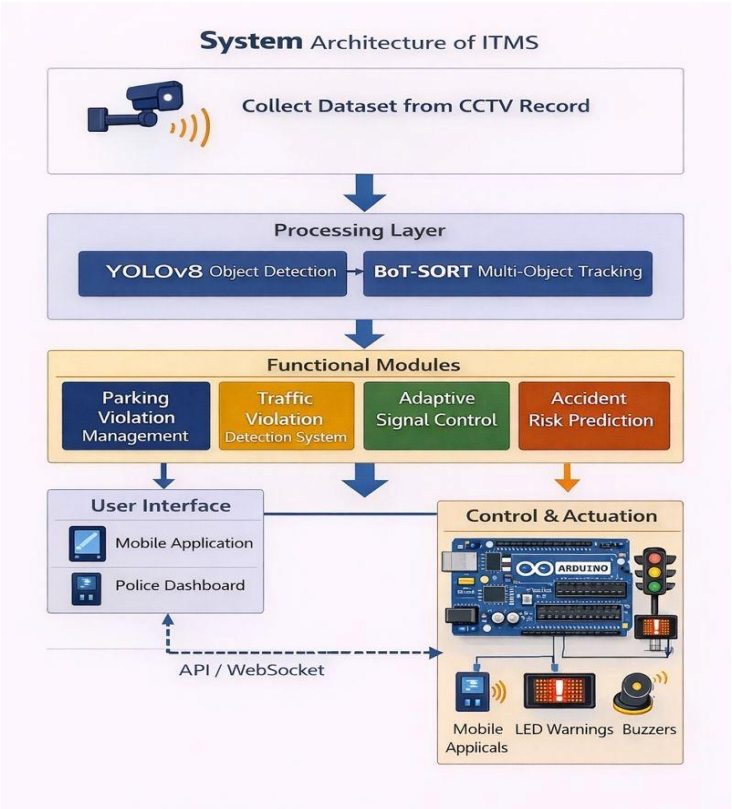

Fig. 1. System architecture of the proposed ITMS.

B. Data Acquisition and Pre-Processing

An integrated data acquisition framework is used in the proposed system to ensure fast and reliable decision-making. High-definition CCTV cameras act as the primary source of visual data, continuously capturing real-time traffic flow and vehicle movements. This visual data allows the system to monitor road conditions and detect different types of vehicles under various environments.

To support the vision data, additional sensors are used to improve system accuracy. Ultrasonic sensors are installed at key intersections to measure vehicle queue lengths aIDongestion levels. RFID technology is also incorporated into detecting approaching emergency vehicles, allowing the system to provide priority signal control when needed. By combining these different data

sources, the system achieves a more accurate and comprehensive understanding of traffic conditions.

The YOLOv8 Deep Learning Model is used to process CCTV video streams in real time to detect and classify vehicles. The model is trained using a localized dataset that includes region-specific vehicle types such as three-wheelers, buses, cars, and motorbikes. This improves detection accuracy in Sri Lankan traffic conditions, especially under different lighting and weather situations.

Before feeding data into the model, several pre-processing steps are applied, including frame extraction, image resizing, normalization, noise reduction, and region of interest (ROI) selection. These steps help improve model performance and reduce computational load.

In dense traffic environments where vehicles may overlap or become partially hidden, the BoT-SORT multi-object tracking algorithm is used to maintain consistent vehicle identities across frames. This ensures stable tracking and accurate vehicle trajectory analysis across all vision-based modules of system. The overall object detection process using YOLOv8 is illustrated in Fig. 2

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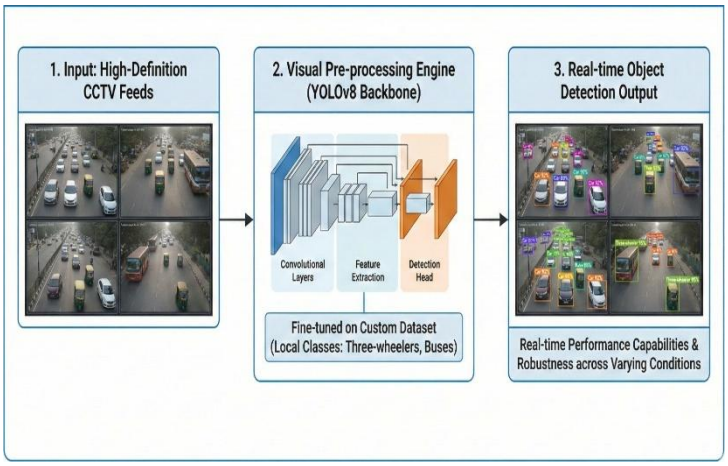

Fig. 2. YOLOv8n Object Detection Process

C. *Core Functional Modules*

The workflow of each functional subsystem is outlined below.

Parking Violation Management: This module counters unauthorized roadside parking using proactive deterrence. Vehicles tracked within restricted zones are monitored for stop duration. A 30-second grace period activates warning LEDs and buzzers to encourage drivers to move. Remaining vehicles are recorded with supporting evidence and trans- mitted to the enforcement dashboard, where dynamic fines are levied based on time and local traffic impact (Fig. 3).

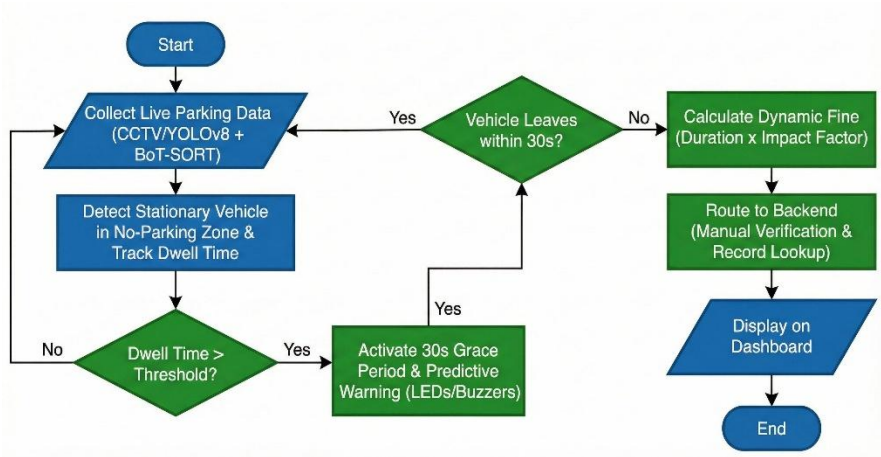

Fig. 3. Parking violation detection flow

Traffic Violation Detection System and Driver Safety Scoreboard: The purpose of this module is to generate an accurate record of video footage that can be reviewed to identify and document traffic violations, including Red Light Violations, Lane Weaving and any other violations/related violations; without documenting duplicate events in one incident. Additionally, every driver who is processed into this system will have a profile created for them that includes their Safety Score (100 at the time of entry). As violations are detected, the Safety Score will begin to decrease according to the severity of the violations detected and the number of violations detected; thus, providing the driver with an overall understanding of their driving behavior. The safety score is designed to reflect a daily recalculated score to ensure the safety score does not fluctuate abruptly and remains consistent over time. As well as backend processing of the system, a mobile application will also be available for all drivers who are processed and will give drivers the ability to view their Safety Score, review their recorded violations and pay any associated fines through the application. Administrative personnel will also have access to a centralized dashboard application (Fig. 4) where they can monitor violations recorded and associated records related to violations, manage evidence collected to support violations, and to conduct all tasks necessary to review cases effectively and efficiently.

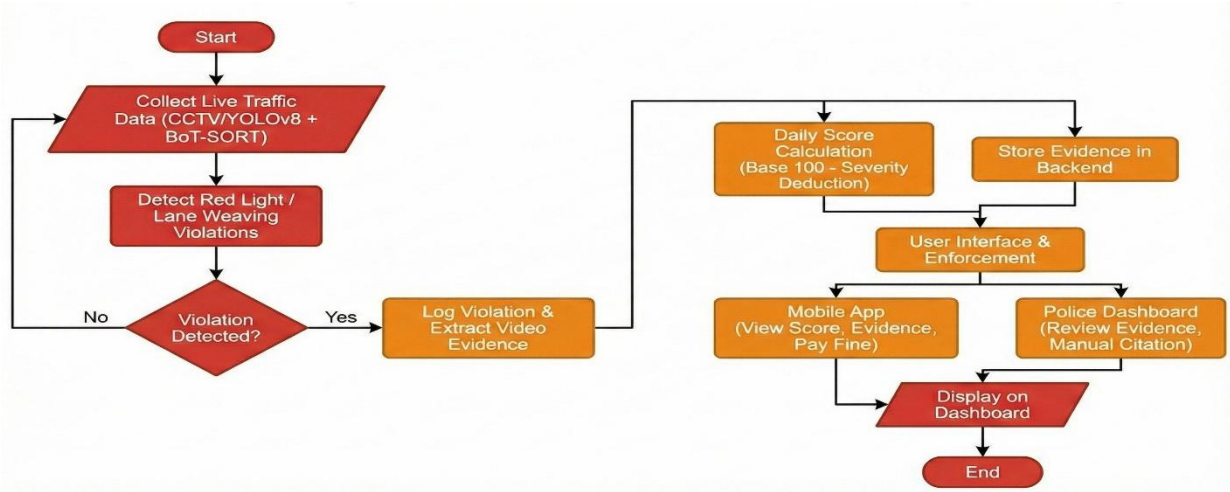

Fig. 4. Traffic violation detection and safety score workflow

Adaptive Signal Control using Fuzzy Logic: To reduce traffic congestion more effectively, the proposed system applies adaptive signal control in which green light durations are adjusted dynamically instead of following fixed and rigid timing schedules [12]. Ultrasonic sensors installed at each lane continuously measure the number of waiting vehicles and estimate the traffic density in real time. Based on these input values, the fuzzy logic controller determines the most suitable signal duration for each direction, allowing lanes with heavier traffic flow to receive longer green times while reducing unnecessary delays for other vehicles. This adaptive approach improves intersection throughput, minimizes idle waiting time, and supports smoother traffic movement under varying road conditions [10]. In addition, the system provides emergency vehicle prioritization through RFID technology. When an approaching emergency vehicle is detected, the controller temporarily overrides the normal signal sequence and safely halts cross-traffic, ensuring a clear and rapid path until the emergency vehicle has com- pletely passed through the junction (Fig. 5).

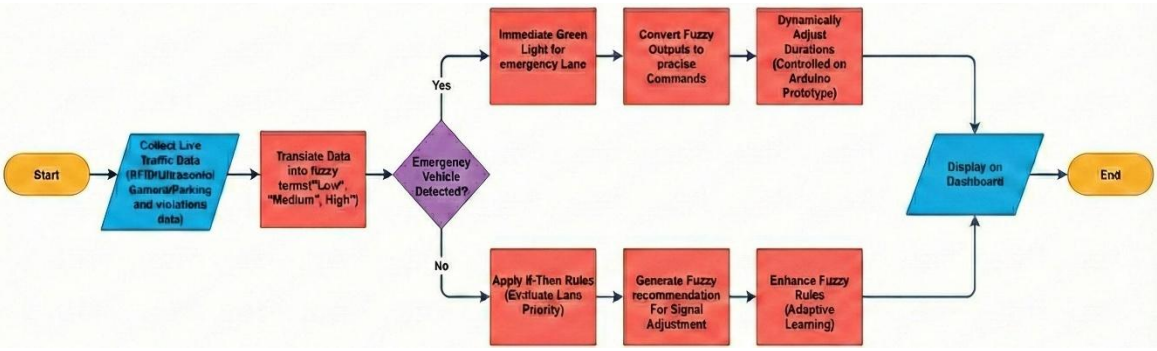

Fig. 5. Adaptive traffic signal control workflow

Accident Risk Prediction: This pipeline classifies high-risk behaviors by computing a weighted dynamic risk score combining an instantaneous Speed Factor with a Violation History Factor

[12]. Scores register as LOW, MEDIUM, HIGH, or CRITICAL. Hazardous events such as sudden braking and lane drifting are also monitored. Risk hotspots are visualized via interactive dashboard heatmaps, triggering automatic alerts upon surpassing critical thresholds to ensure early intervention (Fig. 6).

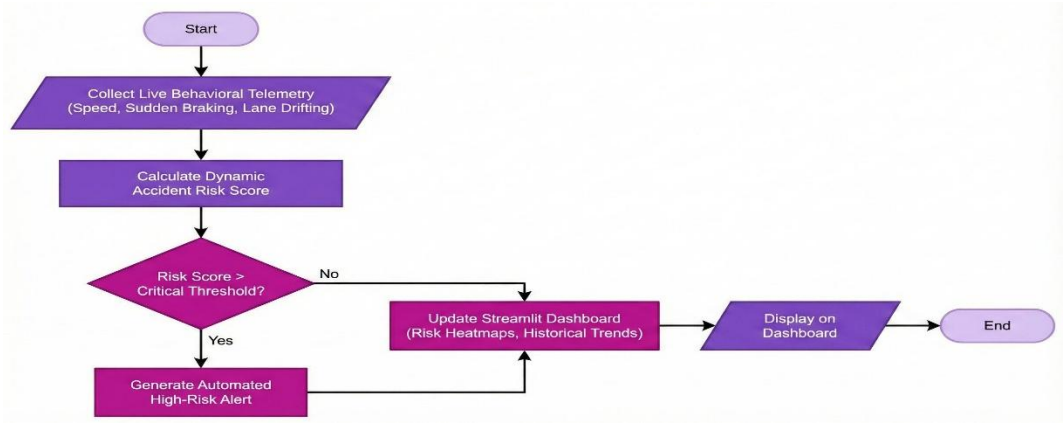

Fig. 6. Accident risk prediction workflow

2.1 Commercialization aspects of the product

The Intelligent Traffic Management System (ITMS) created in this study has tremendous commercialization potential in the Sri Lankan market and other developing countries. The system addresses a vital and expanding demand in metropolitan areas where traffic congestion, unlawful parking, and road accidents are common occurrences. Unlike expensive international traffic management solutions that require huge infrastructure expenditures, the suggested ITMS depends with low-cost components like IR sensors, RFID tags, CCTV cameras, and Arduino microcontrollers. This makes the product inexpensive to local councils, urban development authorities, and private traffic management organizations with minimal budgets. Customers can purchase only the modules they need at first, such as adaptive signal control or parking violation detection, and then add more features as their budget allows.

This product's target market includes local government agencies such as the Colombo Municipal Council, the Sri Lanka Police Department, and the Road Development Authority. These agencies now devote large resources to manual traffic enforcement, which is wasteful and labor-intensive. The ITMS can lower operational expenses by automating violation detection, fine computation, and signal control. The dual-interface ecosystem, which consists of a Command Center Dashboard for authorities and a Driver Mobile Application for residents,

generates several revenue streams. Authorities pay for system installation and dashboard access, and revenue can also be produced by collecting fines via the mobile application, with the system supplier keeping a modest transaction charge.

From a manufacturing and deployment standpoint, the ITMS may be built locally with widely available electronic components and open-source software tools like Python, YOLOv8. This decreases reliance on foreign sources while keeping maintenance costs low. The system can be implemented in stages, beginning with high-priority crossings in Colombo.The use of passive RFID tags for emergency vehicle prioritization does not necessitate constant power or internet connectivity, making the system dependable even in places with intermittent network service. Furthermore, the fuzzy logic controller is efficient and does not require high-performance computing gear, which reduces implementation costs even further. In order to achieve long-term commercialization, the ITMS can be connected with existing government traffic management programs and smart city projects in Sri Lanka. Partnerships with telecommunications companies can enable real-time data transmission using the Message Queuing Telemetry Transport (MQTT) protocol and cloud platforms like Amazon Web Services (AWS). The system can potentially be expanded to provide value-added services such as traffic prediction analytics, insurance company integrations that reward safer drivers with lower premiums, and bus priority for public transportation. With proper intellectual property protection and government policy support for smart traffic solutions, the ITMS has a strong chance of becoming a commercially viable product that improves road safety, reduces congestion, and generates economic value for Sri Lanka, all while being scalable to other South Asian markets facing similar traffic challenges.

2.2 Testing & Implementation

2.2.1 Implementation

The Modular Prototype for Intelligent Traffic Systems was delivered successfully where Computer Vision; IOT Sensors; Fuzzy Logic and Dashboard Based Control were combined to form a complete system. Python was used on the back end, and Open CV and Pytorch were used for processing real time video via CCTV cameras to detect and classify different types of vehicles using YOLO V8n detection algorithms. Vehicle identification was preserved to enable tracking of noticed violations as well as tracking of vehicles that parked.

To provide adaptive control of traffic signal timing, lanes with multiple I.R. or ultrasonic sensors were used to measure the amount of vehicular density in each lane of the intersection that fed into a fuzzy logic controller which classified all lanes as either Low; Medium or High traffic density at an intersection, then made real-time adjustments to the length (duration) of the Green light. Furthermore, RFID technology will be used to provide priority access for emergency vehicles by temporarily overriding the typical cycle timing of the signals

In addition to all of the above, a mobile application and a dashboard were created. The mobile application enabled users to view recorded violation; safety score and fine detail, whereas the dashboard provided authorities with real-time access to live video; violation alerts; near accident risk level and the status of traffic signal control.

2.2.2 Testing

Testing was done to measure the accuracy of detection, how quickly the system could respond, how reliable the sensors were and how easy it was for the user to operate the dashboard. The parking violation detection model using YOLOv8n was evaluated using a parking data set of 945 images (dataset V3) and achieved 79.6% precision, 86.7% recall, 85.6% mAP@0.5 compared to an earlier data set of only 945 images (dataset V2) and performed better than this data set.

A prototype of a traffic signal was tested with IR (infrared) sensors that detected the presence of vehicles in each lane. The fuzzy logic controller was also evaluated to determine if it could correctly adjust the length of time for green lights based on how many vehicles were present in each lane. Finally, RFID-based technology was used to test the detection of emergency vehicles to ensure that when detected by the light control system, emergency vehicles would not have to wait at a red light.

Usability testing was conducted on the dashboard and mobile app used by the command centre for traffic management. Users reported that the dashboard allowed them to effectively manage their areas and made it easier and quicker to monitor multiple locations by combining video feed, alerts for violations, and the ability to control traffic signal timing in one place.

2.2.3 Integration and System Testing

Integration testing occurred to confirm the full integration of all modules of the Intelligent Traffic Management System to function as one large integrated platform. The modules that were tested include the CCTV-Based Vehicle Detection Module, the Parking Violation Detection Module, the

Traffic Violation System, the Adaptive Signal Controller Module, the Accident Risk Prediction Module, the Driver Mobile Application, and the Administrative Dashboard, as all are interconnected in the system.

The system was verified by performing system testing on the ability of the system to process and share data in real-time across modules using the data from the camera/ sensor IoT device. An example of this is that the fuzzy logic controller was able to collect data regarding the density of cars and use this information to change the timing of traffic lights. In addition, the camera response to video based detection allowed for violations to be recorded and reported on the dashboard.

Once the overall system was tested simulating a traffic junction, it was confirmed by the results that it was able to receive data and classify vehicle density using the IoT prototype; it was able to adjust the timing of traffic lights; detect emergency vehicles; provide violation information; and allow for information reporting to support authorities to reach a decision using the dashboard. Therefore, the integration of all modules in the ITMS will aid in the management of urban traffic flow through rapid response and less manual monitoring and support a safer method of managing urban traffic.

3. RESULTS & DISCUSSION

3.1 Results

3.1.1 IoT-Based Traffic Monitoring Results

The IoT-based traffic monitoring module successfully collects and transmits real-time traffic data from distributed sensors. Key parameters such as vehicle density, lane occupancy, and traffic flow patterns are continuously captured and processed to maintain an accurate view of intersection conditions.

The system demonstrates stable performance under different traffic conditions, including low, medium, and high congestion scenarios. Even during peak traffic loads, data transmission remains consistent, ensuring uninterrupted monitoring. This enables the system to quickly detect sudden traffic changes such as congestion buildup or lane blockages and respond accordingly.

3.1.2 Machine Learning Traffic Prediction Results

The machine learning module effectively predicts traffic congestion patterns using historical datasets combined with real-time inputs. The model identifies peak traffic hours and anticipates congestion before it fully develops.

This predictive capability improves system efficiency by enabling proactive adjustments to traffic signal timings. Instead of reacting after congestion occurs, the system prepares in advance, resulting in smoother traffic flow and reduced waiting times.

The results show that predictive modeling significantly contributes to reducing congestion severity during peak traffic conditions.

3.1.3 Adaptive Traffic Signal Control Results

The adaptive traffic signal control system dynamically adjusts signal timings based on real-time traffic density and predictive outputs.

High-density lanes are assigned longer green signal durations, while low-density lanes are processed faster or skipped when necessary. This ensures balanced traffic distribution across all lanes and reduces unnecessary idle waiting time.

Testing results indicate that the adaptive system significantly improves intersection throughput compared to traditional fixed-time signal systems. It is especially effective during peak hours, where congestion is most severe.

3.1.4 Route Optimization System Results

The route optimization module provides real-time alternative route suggestions based on congestion analysis. It helps drivers avoid heavily congested intersections by redirecting them to less busy roads.

Users who follow optimized routes experience reduced travel time and smoother driving conditions. This also contributes to better traffic distribution across the entire road network,

reducing pressure on major junctions. Overall, the module enhances system-wide efficiency and improves user travel experience.

3.1.5 Accident Risk Prediction System Results

The accident risk prediction module is successfully implemented as a core safety component of the system. It uses a dual-factor model combining **vehicle speed analysis and violation history patterns** to calculate real-time risk scores.

The system generates risk levels (Low, Medium, High) based on continuous data evaluation. High-risk situations are identified with strong accuracy and trigger immediate alerts through the integrated dashboard and notification system.

3.2 Research Findings and Discussion

3.2.1 Effectiveness of IoT-Based Traffic Monitoring

The IoT-based monitoring system significantly enhances real-time traffic visibility. Unlike traditional systems, which rely on static or manual data, this system continuously collects live traffic information.

This real-time capability improves decision-making accuracy and ensures faster response to dynamic traffic changes. The study confirms that IoT integration is essential for intelligent and adaptive traffic systems.

3.2.2 Impact of Machine Learning in Traffic Prediction

Machine learning plays a key role in improving traffic prediction accuracy. By analyzing both historical and real-time data, the system can forecast congestion before it fully develops.

This proactive approach allows traffic signals to be adjusted in advance, reducing congestion buildup and improving traffic flow stability.

The findings confirm that predictive analytics significantly improve urban traffic management efficiency.

3.2.3 Effectiveness of Adaptive Traffic Signal Control

The adaptive signal control system improves intersection performance by dynamically adjusting signal timings based on traffic demand.

Unlike fixed-time systems, this approach ensures that road capacity is utilized efficiently. It reduces waiting time, improves vehicle flow, and enhances intersection performance.

The research confirms that adaptive control is more effective in managing modern urban traffic conditions.

3.2.4 Contribution of Route Optimization System

The route optimization module reduces congestion by distributing vehicles across alternative routes. By guiding drivers away from congested intersections, it improves overall network performance and reduces travel time.

The system also helps balance traffic loads across the road network, improving long-term traffic efficiency.

3.2.5 Effectiveness of Accident Risk Prediction System

The accident risk prediction module is one of the most important safety components of the system. It combines **speed behavior analysis and violation history** to evaluate driver risk levels.

The dual-factor model significantly improves prediction accuracy compared to single-variable approaches. It successfully identifies high-risk situations in real time and provides early warnings to prevent potential accidents.

Key findings include:

- Dual-factor model outperforms single-factor models
- Behavioral history is a strong predictor of accident risk
- Real-time risk scoring improves preventive action capability
- Fast alert delivery enhances safety response time

This confirms that behavioral analytics combined with real-time data processing is highly effective in accident risk prevention systems.

3.2.6 Overall System Integration Analysis

The integration of IoT monitoring, machine learning prediction, adaptive signal control, route optimization, and accident risk prediction forms a complete intelligent traffic management ecosystem.

Each module contributes to system efficiency, but their combined operation delivers a significantly stronger performance. Real-time data supports prediction, prediction enhances signal control, and risk analysis improves safety management. The study confirms that integrated intelligent systems are far more effective than isolated traffic solutions.

3.3 Summary of Each Student's contribution

- ☐ **IT22925572 - Gunarathna R.P**

The individual contribution of **IT22925572 – Gunarathna R.P** is focused on the design and implementation of the **Parking Behavior Analysis System**, which is an important analytical module within the Intelligent Traffic Management System (ITMS). The main objective of this component is to monitor, analyze, and evaluate vehicle parking patterns in urban environments in order to reduce congestion, improve road utilization, and support smarter traffic decision-making.

• **System design and requirement analysis of the Parking Behavior Analysis System** was carried out by identifying key parking-related problems in urban traffic environments such as illegal parking, over-parking in restricted zones, and inefficient use of limited parking spaces. The system was designed to address these issues by providing real-time monitoring and behavioral analysis of parking activities.

• **Development of real-time parking data collection module** enables continuous tracking of vehicle entry and exit events in parking zones. The system captures essential data such as vehicle ID, timestamp, location, and duration of stay. This structured data forms the foundation for analyzing parking behavior patterns and detecting anomalies.

• **Implementation of parking behavior analytics engine** focuses on processing collected data to identify meaningful patterns. The system evaluates parking duration, frequency of repeated parking in the same zone, and peak occupancy periods. This allows the system to distinguish between normal and abnormal parking behavior based on predefined thresholds and contextual conditions.

• **Design of parking violation detection mechanism** enables automatic identification of improper parking activities. These include illegal parking in restricted zones, double parking that obstructs traffic flow, and overstaying beyond permitted time limits. Each violation is logged and categorized based on severity, enabling authorities to prioritize enforcement actions.

• **Time-based and location-based analysis module development** allows the system to study parking behavior trends across different time intervals (morning, peak hours, evening) and spatial zones. This helps identify high-demand parking areas and frequently congested zones. The analysis supports better urban planning and efficient allocation of parking infrastructure.

• **Integration of visualization and reporting dashboard** provides real-time insights into parking behavior through interactive charts, heatmaps, and statistical summaries. The dashboard allows traffic administrators to easily identify problematic parking zones and monitor compliance levels. Historical data visualization also supports long-term traffic planning decisions.

• **System integration with ITMS ecosystem** ensures that parking behavior insights contribute to overall traffic management decisions. The module interacts with other components such as route optimization and congestion analysis systems to improve traffic distribution and reduce unnecessary vehicle clustering in specific areas.

• **Performance testing and validation of the system** were conducted using simulated and real-world parking scenarios. The system demonstrated reliable performance in detecting parking violations and analyzing behavioral trends with high consistency. It was able to process continuous data streams without significant delays, making it suitable for real-time deployment.

• **Optimization and scalability considerations** were also addressed during development. The system is designed to handle increasing amounts of parking data as urban areas expand. Efficient

database structuring and modular architecture ensure that the system can be scaled without affecting performance.

Overall, the Parking Behavior Analysis System significantly enhances the Intelligent Traffic Management System by providing deep insights into parking behavior patterns. It helps reduce urban congestion caused by improper parking, improves traffic flow efficiency, and supports better decision-making for city traffic authorities. This contribution plays a crucial role in building a smarter, more organized, and data-driven urban traffic environment.

- ☐ **IT22900890 - Randima K.M.G.D**

Randima K.M.G.D. developed the ITMS's "Smart Traffic Violation Detection" layer. While the team detects vehicles with YOLOv8, this component tracks and analyzes their exact progress through the intersection using the BoT-SORT multi-object tracking method. This tracking feature allows for accurate infraction detection and driver behavior analysis.

- **BoT-SORT Implementation for Chaotic Traffic** – BoT-SORT was implemented to deal with the unpredictable nature of Sri Lankan traffic by ensuring constant vehicle identification between frames. Most trackers lose vehicles under occlusion, however BoT-SORT uses re-identification features to match detected objects to those previously tracked.
- **Reduction of Identity Switches for Lane Weaving** – BoT-SORT keeps vehicle IDs stable even when vehicles pass via narrow gaps. This stability is crucial for lane weaving detection, as ID switching in the middle of a move would prevent the system from detecting a violation.
- **Trajectory Mapping for Lane Weaving Detection** – The center-bottom point of each tracked vehicle was converted to a time-series position set. When the trajectory crosses solid lane divider lines more than twice in 50 meters, a lane weaving violation occurs.
- **Red-Light Synchronization and Entrance Logic** – The exact time a vehicle's monitored ID crosses the stop line when the signal turns red was determined. This interaction between traffic light status and BoT-SORT tracker allows for accurate violation identification at intersections.
- **Automated Evidence Capture for Violations** – When BoT-SORT determines that a vehicle entered the violation zone during a red phase, the module automatically designates that particular ID for evidence collection. A sequence of 3 to 5 frames serves as proof.

- **Driver Safety Score Gamification Component** – Each vehicle tracked by its permanent Identity is assigned a temporary session score beginning at 100 points. Lane weaving deducts 5 points, while red-light jumping deducts 15, ensuring equitable scoring for all drivers.
- **Dynamic and Daily Score Recalculation** – The safety score serves to be adjusted regularly to guarantee consistency throughout time. This avoids quick variations and keeps a dependable indicator of driving risk levels for regulatory purposes.
- **Real-Time Speed Measurement Module** – Vehicle speed gets evaluated in real time by video-based vehicle detection. Over speeding infractions are automatically identified and recorded, along with additional evidence to hold drivers responsible.

☐ **IT22363848 - Tennakoon I.M.S.R**

Tennakoon I.M.S.R. developed and implemented ITMS's core intelligence, which includes a Fuzzy logic controller that processes real-time traffic data and dynamically optimizes signal timing. This component guarantees that intersection operations respond to actual vehicle demand rather than following planned schedules.

- **Data collection and processing entailed gathering real**-time traffic counts from IR sensors, RFID readers, and cameras in order to monitor road conditions. Raw numerical sensor data was converted into fuzzy linguistic variables indicating low, medium, and high traffic flow.
- **Fuzzy Inference Engine Development** - A comprehensive set of IF-THEN rules was developed to determine lane priorities. These rules allow the system to handle fuzzy inputs and make signal adjustments depending on calculated priority ratings.
- **Defuzzification Logic Integration** - Fuzzy recommendations were converted into precise orders for the MCU to change green light timings. Duration settings were set to 5 seconds for low density, 8 seconds for medium density, and 12 seconds for high density.
- **Emergency Vehicle Prioritization Protocol** -The EVP Protocol used RFID technology to prioritize emergency Vehicles in Lane 1. When a valid RFID tag is recognized, the system can set a green light independent of the current traffic flow.
- **Context-Aware Optimization** - Input data from external modules, such as parking and traffic violation detection, was integrated. This guarantees that signal logic remains responsive to the larger context of Sri Lankan traffic situations.

- **Hardware Prototype and Adaptive Learning** - A physical prototype was built with an Arduino to control LED signals and demonstrate the execution of fuzzy logic rules. Adaptive learning capabilities were used to gradually improve the intelligence of the fuzzy rule base over time.

☐ IT22337580 - Palihakkara P.I

The contribution of IT22337580 – Palihakkara P.I focuses on the design, development, and implementation of the **Accident Risk Prediction and Integrated Dashboard** component within the Intelligent Traffic Management System (ITMS). This component plays a critical role in enhancing road safety by enabling real-time monitoring and predictive analysis of potential accident risks.

A major part of the contribution involved the **development of the accident risk prediction model**, which evaluates both real-time vehicle behavior and historical traffic violation data. Key risk factors such as vehicle speed, acceleration patterns, inter-vehicle distance, and abnormal driving behaviors (e.g., sudden braking, lane drifting, and wrong-way movement) were identified and incorporated into a weighted risk scoring mechanism. This model dynamically categorizes vehicles into different risk levels such as low, medium, and high, allowing proactive identification of potentially dangerous situations.

In addition to model development, significant effort was dedicated to the **data processing pipeline**. This included implementing data validation techniques to ensure accuracy and completeness, as well as applying preprocessing methods such as noise reduction and temporal smoothing to improve data reliability. Feature engineering techniques were also applied to derive meaningful indicators such as relative velocity, proximity measures, and behavioral trends, which are essential inputs for the prediction algorithm.

Another key responsibility was the **integration of backend functionalities** to support seamless communication between system components. RESTful APIs were developed to handle data exchange between the data acquisition layer, prediction engine, and database. A structured database system was utilized to store real-time traffic data, predicted risk levels, and historical violation records, ensuring efficient data retrieval and system scalability.

Furthermore, the contribution included the **design and implementation of the integrated dashboard interface**. The dashboard was developed to provide a user-friendly visualization of accident risk insights through charts, status indicators, and real-time updates. This enables traffic authorities and system users to monitor high-risk vehicles and traffic conditions effectively, supporting timely decision-making and intervention.

Testing and evaluation were also carried out to ensure the functionality, accuracy, and performance of the system. Various test cases were designed to validate prediction outputs and dashboard responsiveness under different traffic scenarios. Identified issues were debugged and optimized to enhance overall system reliability.

In summary, the contribution of IT22337580 – Palihakkara P.I significantly supports the objectives of the ITMS by integrating predictive analytics with real-time visualization. This work enables a proactive approach to traffic management, reducing accident risks and improving road safety through intelligent monitoring and data-driven decision-making.

4. CONCLUSION

The increasing complexity of urban traffic systems, particularly in rapidly developing regions such as Sri Lanka, necessitates the adoption of intelligent and adaptive traffic management solutions. Traditional traffic management approaches, which rely heavily on fixed-time signal control and manual monitoring, are no longer sufficient to handle dynamic traffic conditions, rising vehicle density, and growing safety concerns. This research addressed these limitations by designing and implementing an Intelligent Traffic Management System (ITMS) that integrates computer vision, machine learning, and fuzzy logic techniques into a unified, real-time framework.

The primary objective of this study was to develop a system capable of improving traffic flow efficiency, enhancing road safety, and enabling proactive traffic management. To achieve this, four key components were designed and implemented: Parking Violation Detection & Management, Traffic Violation Detection with LiveSafe Score, Fuzzy Logic Adaptive Traffic Signal Controller, and Accident Risk Prediction with an Integrated Dashboard. These components were successfully integrated into a single system architecture, enabling continuous data flow and coordinated decision-making.

The implementation results demonstrate that the proposed system effectively addresses critical traffic management challenges. The parking violation detection module successfully identified illegally parked vehicles and differentiated between temporary stops and actual violations. The dynamic fine calculation mechanism provided a fair and context-aware approach to enforcement. This contributes to reducing congestion caused by improper parking and improves overall traffic discipline.

The traffic violation detection module accurately identified unsafe driving behaviors such as speeding, red-light violations, lane misuse, and wrong-way driving. The introduction of the LiveSafe Score represents a significant contribution of this research, as it provides a real-time safety evaluation of traffic intersections. Unlike traditional systems that focus solely on enforcement, this approach promotes awareness and encourages safer driving behavior by continuously informing road users about current safety conditions.

The fuzzy logic based adaptive traffic signal controller proved to be effective in handling dynamic traffic conditions. By adjusting signal timings based on real-time traffic density and queue length, the system reduced unnecessary delays and improved intersection efficiency. The inclusion of emergency vehicle prioritization further enhances the practical applicability of the system, ensuring that critical services receive immediate right-of-way.

The accident risk prediction module introduced a proactive dimension to traffic management. By analyzing vehicle speed patterns and violation history, the system was able to classify vehicles into different risk levels and identify potentially dangerous situations in advance. This predictive capability shifts the focus from reactive to preventive traffic management, which is essential for reducing accident rates and improving road safety.

The integrated dashboard played a crucial role in consolidating system outputs and presenting them in a clear and accessible manner. By providing real-time visualization of traffic conditions, violations, safety scores, and risk levels, the dashboard enhances situational awareness and supports data-driven decision-making for traffic authorities.

Overall, the system achieved satisfactory performance across all evaluation metrics. High detection accuracy, low processing latency, and reliable system integration demonstrate the feasibility of implementing such an intelligent system in real-world environments. Compared to

conventional traffic management approaches, the proposed ITMS offers significant improvements in responsiveness, efficiency, and safety monitoring.

Despite these achievements, certain limitations were identified. The performance of the system is influenced by the quality of video input, environmental conditions such as lighting and weather, and computational resource availability. Additionally, the system was evaluated primarily using simulated or controlled datasets, which may not fully represent complex real-world traffic scenarios.

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